ITNOW: GDP level and growth forecasts

Methodological note

Let Y_q denote GDP in chain-linked terms in quarter q. We use ISTAT's seasonally and calendar adjusted series, with reference year 2020, in millions of euros. The subscript q always refers to the target quarter. For example, our January and February nowcasts both refer to Q1.

Log growth and level. Our bridge equation forecasts quarterly log growth,

$$y_q = \log Y_q - \log Y_{q-1}.$$

We estimate the equation directly in these units. We then obtain the GDP level as

$$\widehat{Y}_q = Y_{q-1}\exp(\widehat{y}_q), \qquad \widehat{Y}_q^{\text{billions}} = \frac{\widehat{Y}_q}{1000}.$$

We enter $\widehat{y}_q$ as a decimal, namely log growth of 1% means $\widehat{y}_q = 0.01$.

Quarter-on-quarter and year-on-year growth. We convert the log-growth nowcast into a quarter-on-quarter (QoQ) percentage change, while for year-on-year (YoY) growth we compare the nowcast level with GDP four quarters earlier:

$$\widehat{r}_q^{\text{QoQ}} = 100\big[\exp(\widehat{y}_q) - 1\big], \qquad \widehat{r}_q^{\text{YoY}} = 100\left(\frac{\widehat{Y}_q}{Y_{q-4}} - 1\right).$$

Annual carry-over growth. Carry-over answers a simple question: what would annual GDP growth be if GDP stayed at our nowcast level for the rest of the year?

We write $Y_{i,k}$ for GDP in quarter k of year i, and suppose we are nowcasting quarter j. The annual average is

$$\widehat{\overline{Y}}_i^{\text{flat}} = \frac{\displaystyle\sum_{k=1}^{j-1} Y_{i,k} + (5-j)\widehat{Y}_{i,j}}{4}.$$

The factor $5 - j$ counts how many quarters use the nowcast level, including quarter j itself. We compare this with last year's average GDP level:

$$\widehat{c}_{i,j} = 100\left(\frac{\widehat{\overline{Y}}_i^{\text{flat}}}{\overline{Y}_{i-1}} - 1\right).$$

where $\overline{Y}_{i-1} = \frac{1}{4}\sum_{k=1}^{4} Y_{i-1,k}$.

For example, suppose last year's quarterly levels were all 100. If this year's Q1 is 101 and we nowcast Q2 at 102, then we compute the average of the sequence $(101, 102, 102, 102)$, equal to 101.75, and the carry-over is 1.75%.

GDP measure and carry-over definition: ISTAT, Quarterly national accounts, Q1 2026, glossary (p. 4).